

Defective Analysis and the $\lambda\rho$ Derivation of COTG

A unified formal draft: clocks, algebraic spine, retained kernels, $\lambda\rho$ differentials, and differential-geometric COTG

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Claim boundary. This document is a formal synthesis of the uploaded Defective Analysis drafts and the current alignment corrections. It treats θ and α as *clocks*: indices over admissible operation modes. It defines a rigorous presentation category in which a bijection/equivalence can be stated between the algebraic spine, DA-derived COTG, and differential-geometric COTG. It does *not* assert an externally peer-reviewed theorem of standard mathematics independent of these declared hypotheses. Where a universal claim is still unproved, it is marked as a hypothesis or theorem-candidate.

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1 Executive compression

The corrected primitive is not

$$\theta = \text{projection}, \quad \alpha = \text{hidden support}.$$

That is type-wrong. The corrected primitive is:

$$\boxed{\theta, \alpha \text{ are clocks: indices over admissible modes of operation.}}$$

The θ -clock indexes exposure operations: delete, differentiate, project, probe, compare, or coarse-grain. The α -clock indexes reconstructive operations: glue, integrate, complete, realize, evolve, or reconstruct.

Defective Analysis is the clock-assignment and reconstruction grammar:

$$\text{DA}(P) = (\text{Clk}(P), D_\theta, I_\alpha, F_\theta, K_{\text{ret}}, \Pi, \sigma, \text{Adm}).$$

Here D_θ and I_α are operations generated under clocks. They are not the clocks themselves.

The algebraic spine is

$$0 \rightarrow \{0, 1\} \rightarrow \mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C},$$

compressed as

$$\boxed{\mathbb{C} = S_{\min} + \theta^\pm + \alpha^\pm + \lambda + \rho.}$$

Here λ is the invariant-completion obstruction/repair and ρ is the polar-complexification obstruction/repair. In local differential form, the nonzero complex mode admits the exact polar decomposition

$$z = r e^{i\varphi}, \quad d \log z = d \log r + i d\varphi.$$

This is the rigorous seed of the $\lambda\rho$ differential:

$$\boxed{A_{\lambda\rho} = A_\lambda + i A_\rho.}$$

A_λ is the magnitude/completion differential; A_ρ is the phase/polar differential. The full local symmetry is

$$\boxed{G_{\lambda\rho} = \mathbb{C}^\times \rtimes \mathbb{Z}_2^\rho \cong (\mathbb{R}_{>0} \times \text{U}(1)) \rtimes \mathbb{Z}_2^\rho,}$$

where the $\mathbb{Z}_2^\rho$ element is the polar/Moebius inversion, e.g. complex conjugation or orientation reversal.

COTG is the charted oscillatory transport realization of DA/WDS data:

$$\text{COTG} = (M, \mathcal{A}, E, \{X_i\}, \{q_i\}, \{g_{ij}\}, T, \Pi, R).$$

Its non-metaphorical differential geometry is a $G_{\lambda\rho}$ -bundle/associated complex vector-bundle geometry equipped with local sections, transition maps, $\lambda\rho$ connection, curvature, holonomy, projection, and residuals.

The desired equivalence is valid under a restricted, declared presentation category:

$$\boxed{\text{Sp}_{\lambda\rho} \simeq \text{DA-COTG} \simeq \text{DG-COTG}.}$$

This is not a vague analogy. It is a typed equivalence between three descriptions of the same finite, non-leaking, clock-declared, $\lambda\rho$ -regular object.

2 Source conventions and non-negotiable correction

2.1 Clock-index doctrine

Definition 2.1 (Clock). *A clock is an index over an admissible mode of operation. A clock is not ordinary time, not a coordinate, not a dimension, and not a carrier. A clock determines which operations are legal in a given problem description.*

Definition 2.2 (θ -clock and α -clock). *The θ -clock indexes operations that expose or reduce data: projection, derivation, deletion, probing, comparison, and coarse-graining.*

The α -clock indexes operations that reconstruct or realize data: integration, gluing, completion, realization, evolution, and reconstruction.

Remark 2.3 (Type discipline). *It is illegal in this grammar to say “ θ is projection” or “ α is hidden support” without qualification. The legal form is:*

D_θ is a defective observable generated under the θ -clock,

I_α is a reconstruction operation generated under the α -clock.

Similarly, hidden support is not the α -clock. Hidden support is carrier/fiber/kernel data that become legally accessible only through declared α -clock operations.

3 Defective Analysis

3.1 Problem descriptions

Definition 3.1 (DA problem description). *A Defective Analysis problem description is a tuple*

$$\mathfrak{P} = (P, X, \sim, \text{Clk}, D_\theta, I_\alpha, K, \Pi, \sigma, \text{Adm})$$

where:

- P is the problem being described;
- X is the carrier class of admissible objects or states;
- $\sim$ is the declared reconstruction equivalence;
- $\text{Clk}(P) \subseteq \{\theta, \alpha\}$ is the clock signature;
- $D_\theta : X \rightarrow Y$ is the defective observable generated under the θ -clock;
- $I_\alpha : Y \times K \rightarrow X / \sim$ is the reconstruction/realization operation generated under the α -clock;
- K is the candidate retained-data class;
- Π is a visible quotient or comparison map;
- $\sigma : Y \rightarrow \Sigma \cup \{\perp\}$ is the polarity classifier;
- Adm is the admissibility predicate.

Definition 3.2 (Clock signature).

$$\text{Clk}(P) \subseteq \{\theta, \alpha\}.$$

A problem is θ -local if $\text{Clk}(P) = \{\theta\}$, α -local if $\text{Clk}(P) = \{\alpha\}$, and $\theta\alpha$ -global if $\text{Clk}(P) = \{\theta, \alpha\}$. A problem is malformed when the assigned clock signature does not match the operations required for well-formedness.

Definition 3.3 (Defective fiber). For $y \in Y$,

$$F_\theta(y) = D_\theta^{-1}(y) = \{X \in \mathcal{X} : D_\theta X = y\}.$$

For $X \in \mathcal{X}$,

$$F_\theta(X) = D_\theta^{-1}(D_\theta X).$$

This is the candidate class sharing the same defective observable.

Definition 3.4 (Retained kernel).

$$K_{\text{ret}}(X) = \{k \in K : I_\alpha(D_\theta X, k) \sim X \text{ and } \text{Adm}(X, k)\}.$$

The retained kernel is admissible selection data. It is not the entire fiber unless that identification is separately proved.

Definition 3.5 (Flat and complete reconstruction). If $0 \in K$ is the null retained datum, flat reconstruction is

$$I_\alpha^0 D_\theta(X) = I_\alpha(D_\theta X, 0).$$

Complete reconstruction is any reconstruction

$$I_\alpha(D_\theta X, k), \quad k \in K_{\text{ret}}(X).$$

A nontrivial defect is present when

$$I_\alpha^0 D_\theta(X) \not\sim X.$$

Definition 3.6 (Clock-order residual). A clock-order residual is a typed comparison

$$\Delta_{\theta, \alpha}(X; k) = \text{cmp}(D_\theta I_\alpha(D_\theta X, k), D_\theta X) \in R,$$

where R is a declared residual carrier and cmp is a declared comparison operation. This notation is legal only when both terms have been typed into a common comparison carrier.

Definition 3.7 (No-leakage). A DA argument satisfies no-leakage when every quantity used in the solution is either supplied in the data or generated by a declared legal operation. If a quantity identifies a hidden representative, hidden support, or missing reconstruction choice without legal data, the argument leaks.

Equivalently, if $k \sim_{\text{hid}} k'$ is killed by a visible quotient, then no declared downstream visible composite may separate k from k' unless that separation is declared as a legal probe.

3.2 The DA object

The compressed object of Defective Analysis is

$$\boxed{\text{DA}(P) = (\text{Clk}(P), D_\theta, I_\alpha, F_\theta, K_{\text{ret}}, \Pi, \sigma, \text{Adm}).}$$

This is the top-level grammar. Defective Calculus begins once the projection/reconstruction structure has been declared.

4 The algebraic spine

4.1 Generator seed

Definition 4.1 (Minimal generator seed).

$$S_{\min} = \{0, 1\}.$$

The element 0 is additive identity. The element 1 is the unit generator. The element 2 is not primitive; it is generated by $2 = 1 + 1$.

4.2 Successive repairs

The scalar ladder is a sequence of typed repairs:

$$0 \rightarrow \{0, 1\} \rightarrow \mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}.$$

The compact clock/obstruction notation is:

$$\begin{aligned} \mathbb{N} &= (\theta^+, \alpha^+), & \mathbb{Z} &= (\theta^\pm, \alpha^+), \\ \mathbb{Q} &= (\theta^\pm, \alpha^\pm), & \mathbb{R} &= (\theta^\pm, \alpha^\pm, \lambda), \\ \mathbb{C} &= (\theta^\pm, \alpha^\pm, \lambda, \rho). \end{aligned}$$

Here θ and α are still clocks. In the scalar ladder specifically, the θ -clock indexes additive successor/inverse-successor operations; the α -clock indexes multiplicative aggregation/inverse-aggregation operations.

Definition 4.2 (λ obstruction). *λ is the obstruction exposed when presentation load is mistaken for invariant scalar size. The repair is to pass to gauge-invariant size. On $\mathbb{Q}$, invariant sizes are absolute values; generator visibility selects the Archimedean branch; completion forces $\mathbb{R}$.*

Definition 4.3 (ρ obstruction). *ρ is the polarization obstruction exposed by real exponentiation: real scaling alone does not encode magnitude-preserving rotation. The repair is polar complexification, allowing*

$$e^{it} = \cos t + i \sin t, \quad |e^{it}| = 1.$$

Proposition 4.4 (Algebraic spine compression). *Under the declared scalar-ladder reading,*

$$\boxed{\mathbb{C} = S_{\min} + \theta^\pm + \alpha^\pm + \lambda + \rho.}$$

Expanded: $\mathbb{C}$ is obtained from the minimal generator seed by adjoining additive inverse closure, multiplicative inverse closure, gauge-invariant Archimedean completion, and polar complexification.

Proof. The construction follows the standard stages: $\mathbb{N}$ is generated by successor from $S_{\min}$; $\mathbb{Z}$ is group completion of $\mathbb{N}$; $\mathbb{Q}$ is localization of $\mathbb{Z}$ at nonzero elements; $\mathbb{R}$ is completion under the selected invariant Archimedean uniformity; $\mathbb{C}$ adjoins a polar unit i with $i^2 = -1$. The compact expression records the clock and obstruction labels attached to these standard repairs. $\square$

5 From algebraic spine to $\lambda\rho$ differentials

5.1 Polar logarithmic differential

The nonzero complex scalar has the unique polar form

$$z = re^{i\varphi}, \quad r \in \mathbb{R}_{>0}, \quad \varphi \in \mathbb{R}/2\pi\mathbb{Z}.$$

Taking logarithmic differential gives

$$\boxed{d \log z = d \log r + i d \varphi.}$$

This is the local non-metaphorical bridge from the algebraic spine to differential geometry.

Definition 5.1 ($\lambda\rho$ differential). *A $\lambda\rho$ differential on a local nonzero mode is*

$$A_{\lambda\rho} = A_\lambda + iA_\rho,$$

where A_λ is a real magnitude/completion 1-form and A_ρ is a real phase/polar 1-form. Locally, for a nonzero scalar mode z ,

$$A_\lambda = d \log |z|, \quad A_\rho = d \arg z.$$

Remark 5.2 (Why this is not metaphor). *The decomposition $\mathbb{C}^\times \cong \mathbb{R}_{>0} \times \mathrm{U}(1)$ is a standard Lie group decomposition. The differential $d \log z = d \log |z| + i d \arg z$ is a standard differential identity on the nonzero complex line, modulo branch choices for argument. The DA reading is not inventing this identity. It assigns it a role: λ controls invariant magnitude/completion, while ρ controls phase/polar orientation.*

5.2 Symmetry group

Definition 5.3 ($\lambda\rho$ symmetry group). *The local nonzero $\lambda\rho$ transport symmetry is*

$$G_{\lambda\rho}^0 = \mathbb{C}^\times \cong \mathbb{R}_{>0} \times \mathrm{U}(1).$$

The Moebius/polar inversion extension is

$$\boxed{G_{\lambda\rho} = \mathbb{C}^\times \rtimes \mathbb{Z}_2^\rho.}$$

The nontrivial element of $\mathbb{Z}_2^\rho$ acts by an orientation-reversing polar operation, for example

$$\tau(z) = \bar{z}$$

or, in a normalized reciprocal convention,

$$\tau(z) = \frac{1}{\bar{z}}.$$

The convention must be declared per realization.

Proposition 5.4 (Moebius non-tautology condition). *A self-return loop is non-tautological when its monodromy lies in the nonidentity component of $G_{\lambda\rho}$:*

$$\mathrm{Hol}_{\lambda\rho}(\gamma) \in \mathbb{C}^\times \tau, \quad \tau^2 = 1, \quad \tau \neq 1.$$

Then one circuit returns an inverted local chart; two circuits recover orientation.

Proof. If the return map is identity, the loop gives no typed distinction. If the return map is τ , then the returned local representative is related to the original by an orientation-reversing involution. Since $\tau^2 = 1$, two traversals restore the original orientation. Thus self-reference is not flat identity; it is twisted closure. $\square$

6 Whole Defect Systems and DA-derived COTG

Definition 6.1 (Whole Defect System). *A Whole Defect System is a tuple*

$$\mathcal{N} = (\mathcal{A}_0, X, \Delta, \{K_p\}_{p \in P}, G, q, \text{Adm}_{\mathcal{N}})$$

where $\mathcal{A}_0$ is the ambient category or comparison setting, X is the pre-quotient carrier, Δ is the defect extractor, K_p are defect sectors, G is the gluing/comparison carrier, q is the quotient or gluing projection, and $\text{Adm}_{\mathcal{N}}$ is the admissibility predicate.

Definition 6.2 (Primal derivative and primal integral). *The Primal derivative is defect extraction:*

$$D_P(\mathcal{N}) = \Delta\mathcal{N}.$$

The Primal integral is partial reconstruction from quotient or gluing data:

$$I_P(\text{quotient data}) = \mathcal{N} \bmod K_{\text{killed}}.$$

It reconstructs a unique pre-quotient WDS only if the retained kernel is supplied.

Definition 6.3 (α -defect equivalence class). *Fix an observable $y \in Y$ and retained data K . Define*

$$k \equiv_{\alpha} k'$$

when k and k' are indistinguishable by every declared visible composite and reconstruct the same target modulo $\sim$ under the α -clock:

$$I_{\alpha}(y, k) \sim I_{\alpha}(y, k'), \quad \Pi(k) = \Pi(k'),$$

with all downstream visible composites also equal. The α -defect equivalence class is

$$K_{\alpha}(y) = [k]_{\equiv_{\alpha}}.$$

Definition 6.4 (DA-derived finite COTG realization). *A finite DA-derived COTG realization is a tuple*

$$\mathcal{C} = (M, \mathcal{A}, E, \{X_i\}_{i \in I}, \{q_i\}_{i \in I}, \{g_{ij}\}_{(i,j) \in E}, T, \Pi, R).$$

Here:

<i>Symbol</i>	<i>Meaning</i>
M	base chart domain or index space
$\mathcal{A} = \{(U_i, \varphi_i)\}$	chart family / atlas
E	directed overlap edge set
X_i	local complex state space or fiber over U_i
q_i	local oscillatory section
g_{ij}	transition map on an overlap $U_i \cap U_j$
T	global tensor or completed state object built from local fibers
Π	projection / measurement map into observable space
R	residual data for gluing, compatibility, or projection failure

Definition 6.5 (Local oscillator). *On a chart U_i , let X_i be a finite-dimensional complex vector space with basis $\{e_{i1}, \dots, e_{iN_i}\}$. A local oscillatory section is*

$$q_i(t) = \sum_{k=1}^{N_i} a_{ik} \exp(i(\omega_{ik}t + \phi_{ik})) e_{ik}.$$

The coefficients determine latent magnitude; the phase terms determine relative transport and projected interference.

Definition 6.6 (Off-axis α displacement and ρ phase transport). *An off-axis α -coordinate in chart U_i determines a displacement*

$$\Delta t_i = \lambda_i \alpha_i.$$

Substitution gives

$$q_i(t + \lambda_i \alpha_i) = \sum_k a_{ik} \exp(i(\omega_{ik}t + \phi_{ik})) \exp(i\omega_{ik}\lambda_i \alpha_i) e_{ik}.$$

The factor $\exp(i\omega_{ik}\lambda_i \alpha_i)$ is the local ρ -phase transport induced by α -clock displacement through the λ scale coefficient.

Lemma 6.7 (Local magnitude invariance). *Off-axis α displacement by ρ -phase transport preserves each local mode magnitude.*

Proof. Each coefficient is multiplied by $\exp(i\vartheta)$ for some real ϑ . Since $|\exp(i\vartheta)| = 1$, the magnitude is unchanged. $\square$

7 Differential-geometric COTG

7.1 COTG as a bundle geometry

Definition 7.1 (Differential-geometric COTG). *A differential-geometric COTG object is a tuple*

$$\mathcal{C}_{\text{dg}} = (M, E_{\lambda\rho}, G_{\lambda\rho}, \nabla_{\lambda\rho}, \Pi, R, \text{Adm})$$

where:

- M is a smooth or finite-smooth charted base;
- $E_{\lambda\rho} \rightarrow M$ is a complex vector bundle or finite charted replacement object;
- $G_{\lambda\rho} = \mathbb{C}^\times \rtimes \mathbb{Z}_2^\rho$ is the scale-phase-inversion structure group, or a matrix representation of it;
- $\nabla_{\lambda\rho} = d + A_{\lambda\rho}$ is a connection with $A_{\lambda\rho} = A_\lambda + iA_\rho$ locally;
- $\Pi : E_{\lambda\rho} \rightarrow Y$ is the observable projection;
- R records curvature, gluing, projection, and reconstruction residuals;
- Adm records admissibility/no-leakage.

Definition 7.2 (Transition maps). *On $U_{ij} = U_i \cap U_j$, local trivializations are related by*

$$q_j = g_{ij} q_i, \quad g_{ij} : U_{ij} \rightarrow G_{\lambda\rho}.$$

For the orientation-preserving component,

$$g_{ij} = \exp(u_{ij} + i\chi_{ij}),$$

with $u_{ij} : U_{ij} \rightarrow \mathbb{R}$ and $\chi_{ij} : U_{ij} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$. The u_{ij} part is the λ transition; the χ_{ij} part is the ρ transition. A Moebius transition includes the τ component.

Definition 7.3 ($\lambda\rho$ connection and curvature). *The local $\lambda\rho$ connection is*

$$A_i = A_i^\lambda + iA_i^\rho.$$

In the abelian line case, the curvature is

$$F_i = dA_i = dA_i^\lambda + i dA_i^\rho.$$

In the matrix/vector case,

$$F_i = dA_i + A_i \wedge A_i.$$

Definition 7.4 (COTG residual). *The COTG residual package is*

$$R = (R_{\text{glue}}, R_{\text{curv}}, R_{\Pi}, R_{\text{rec}}),$$

where:

- R_{glue} measures failure of transition cocycles;
- R_{curv} is curvature or holonomy obstruction;
- R_{Π} measures the lost projection kernel or interference-sensitive quotient;
- R_{rec} measures failure of flat reconstruction.

Proposition 7.5 (Differential geometry resulting from λ and ρ). *The $\lambda\rho$ data induce a genuine differential geometry: a $G_{\lambda\rho}$ -bundle/associated-vector-bundle geometry with connection, curvature, holonomy, projection, and residuals.*

Proof. The pair (A_λ, A_ρ) defines a local connection $A_{\lambda\rho} = A_\lambda + iA_\rho$ valued in the Lie algebra of $\mathbb{C}^\times$ or its matrix representation. Transition maps $g_{ij} : U_{ij} \rightarrow G_{\lambda\rho}$ define bundle gluing. Curvature is computed by $F = dA + A \wedge A$. Holonomy is computed by path-ordered exponentiation. Projection Π maps the associated fiber geometry into observable data. These are the standard ingredients of differential geometry. The DA content is the typed interpretation: λ is magnitude/completion data, ρ is phase/polar data, and R records what fails or is killed under projection/reconstruction. $\square$

7.2 Projection and interference

Let the global tensor state be

$$T = \bigotimes_{i \in I} X_i, \quad Q(t; \alpha) = \bigotimes_{i \in I} q_i(t + \lambda_i \alpha_i).$$

Let

$$y(t; \alpha) = \Pi Q(t; \alpha).$$

Projection may expose relative phase through interference even when local phase transport preserves magnitude.

For a two-sector probe, write

$$z(\phi) = z_+ + e^{i\phi} z_-.$$

The scalar response under a transport operator A is

$$R_\phi(k) = z(\phi)^* A^k z(\phi),$$

so

$$R_\phi(k) = z_+^* A^k z_+ + z_-^* A^k z_- + e^{i\phi} z_+^* A^k z_- + e^{-i\phi} z_-^* A^k z_+.$$

A global phase cancels, but a relative phase can change the projected scalar response. Phase sensitivity is detected when

$$\frac{\partial R_\phi(k)}{\partial \phi} \neq 0$$

for some legal k .

8 The explicit three-way equivalence

8.1 Presentation categories

Definition 8.1 (Algebraic $\lambda\rho$ spine category). *Let $\mathbf{Sp}_{\lambda\rho}$ be the category whose objects are finite presentations*

$$S = (S_{\min}, \theta^\pm, \alpha^\pm, \lambda, \rho, K_\alpha, \text{Adm}_S)$$

where θ, α are clocks, λ is a declared invariant-completion datum, ρ is a declared polar datum, K_α is a defect equivalence class, and Adm_S is an admissibility rule. Morphisms preserve clocks, generator seed, λ data, ρ data, and defect equivalence classes up to declared gauge.

Definition 8.2 (DA-COTG category). *Let $\mathbf{DA-COTG}$ be the category whose objects are DA-derived COTG realizations*

$$\mathcal{C} = (M, \mathcal{A}, E, \{X_i\}, \{q_i\}, \{g_{ij}\}, T, \Pi, R, \text{Adm})$$

obtained from WDS data by admissible quotient/realization. Morphisms preserve chart overlaps, local fibers, transition maps, tensor carriers, projections, residual laws, and admissibility.

Definition 8.3 (Differential-geometric COTG category). *Let $\mathbf{DG-COTG}$ be the category whose objects are differential-geometric COTG packages*

$$(M, E_{\lambda\rho}, G_{\lambda\rho}, \nabla_{\lambda\rho}, \Pi, R, \text{Adm}).$$

Morphisms are bundle maps preserving $G_{\lambda\rho}$ -structure, connection up to gauge, projection, residual, and admissibility.

8.2 The three maps

Definition 8.4 ($\text{Map } F : \mathbf{Sp}_{\lambda\rho} \rightarrow \mathbf{DA-COTG}$). *The functor F sends:*

<i>Algebraic-spine datum</i>	<i>DA-COTG datum</i>
$S_{\min}$	local generator modes / basis seed
θ -clock	allowed D_θ exposure/probe operations
α -clock	allowed I_α gluing/reconstruction operations
λ	invariant magnitude scale and displacement coefficients λ_i
ρ	phase/polar transport and inversion data
K_α	retained defect equivalence class / selection data
Adm_S	no-leakage and reconstruction admissibility

The resulting object is the DA-derived finite COTG presentation.

Definition 8.5 (Map $G : \mathbf{DA-COTG} \rightarrow \mathbf{DG-COTG}$). The functor G sends the chart family, local complex fibers, transition maps, and oscillatory sections into a $G_{\lambda\rho}$ -bundle with local connection

$$A_i = A_i^\lambda + iA_i^\rho,$$

where $A_i^\lambda = d \log |q_i|$ on nonzero modes and $A_i^\rho = d \arg(q_i)$ plus declared α -clock phase displacement. The projection Π and residual R are preserved.

Definition 8.6 (Map $H : \mathbf{DG-COTG} \rightarrow \mathbf{Sp}_{\lambda\rho}$). The functor H recovers:

- a generator seed from local mode bases or declared generating sections;
- θ and α clock signatures from declared projection and reconstruction operation classes;
- λ from the magnitude part of the connection/transition data;
- ρ from the phase/polar part and the inversion component of holonomy;
- K_α from the retained kernel/equivalence class required for admissible reconstruction.

This recovery is valid only in the clock-declared subcategory.

8.3 Presentation equivalence theorem

Theorem 8.7 ($\lambda\rho$ COTG presentation equivalence). Restrict to finite, clock-declared, non-leaking, $\lambda\rho$ -regular presentations in which:

- (i) every local mode used for transport is nonzero on its chart or has a declared singularity type;
- (ii) transition maps land in $G_{\lambda\rho}$;
- (iii) all projection kernels and retained kernels are declared;
- (iv) no quotient is used without a no-leakage proof;
- (v) clock signatures are preserved by morphisms;
- (vi) gauge equivalences identify only changes of local trivialization, not changes of clock signature or killed fiber.

Then there is an equivalence of presentations

$$\boxed{\mathbf{Sp}_{\lambda\rho} \simeq \mathbf{DA-COTG} \simeq \mathbf{DG-COTG}.$$

Proof. Construct F, G, H as above.

First, $G \circ F$ takes algebraic data $S_{\min}, \theta^\pm, \alpha^\pm, \lambda, \rho, K_\alpha$ to a DA-COTG chart presentation and then to a differential-geometric package. The seed determines local basis modes; the clocks determine legal projection and reconstruction operation classes; λ and ρ determine the magnitude and phase parts of local transition/connection data; K_α determines retained equivalence data. Thus $G \circ F$ is defined on every admissible algebraic object.

Second, $H \circ G \circ F$ recovers the original algebraic data up to declared gauge. The nonzero complex polar decomposition is unique modulo the 2π phase branch and gauge choice:

$$z = re^{i\varphi} \iff (r, \varphi \bmod 2\pi).$$

Thus λ and ρ are recovered from $d \log r$ and $d\varphi$. Clock signatures are recovered because the subcategory requires them to be declared and preserved. The retained equivalence class is recovered because no-leakage forbids hidden killed data from changing visible composites.

Third, take any differential-geometric COTG object in the declared subcategory. Its $G_{\lambda\rho}$ -transition maps decompose locally into magnitude, phase, and inversion components. These produce λ , ρ , and Moebius/polar data. The declared projection/reconstruction operation classes recover the θ - and α -clock signatures. The kernel of projection and reconstruction residual recover K_α up to the declared retained equivalence. Therefore $F \circ H$ reconstructs a DA-COTG presentation isomorphic to the original object.

The equivalence is therefore not between arbitrary mathematical objects. It is between three presentations of the same finite, declared, admissible, $\lambda\rho$ -regular COTG structure. $\square$

Remark 8.8 (Meaning of bijection). *The word “bijection” here means equivalence of typed presentation categories after quotienting by declared gauge. It does not mean a naive bijection between θ -space and α -space. The working hypothesis says no naive bijection from θ -space to α -space exists. Representation grammar supplies a tangent equivalence between descriptions, not identity of clocks or carriers.*

9 Full symmetry of COTG

9.1 Local and global symmetry

The local symmetry of a nonzero mode is

$$\mathbb{C}^\times = \mathbb{R}_{>0} \times \mathrm{U}(1).$$

The extended polar/Moebius symmetry is

$$G_{\lambda\rho} = \mathbb{C}^\times \rtimes \mathbb{Z}_2^\rho.$$

For n independent modes, the diagonal symmetry is

$$(G_{\lambda\rho})^n.$$

For coupled modes, one may use a matrix representation

$$G_{\lambda\rho} \hookrightarrow GL(n, \mathbb{C}) \rtimes \mathbb{Z}_2^\rho,$$

with the $\mathbb{Z}_2^\rho$ action implemented by conjugation or a declared orientation-reversing involution.

9.2 Gauge transformation law

A local gauge transformation $h_i : U_i \rightarrow G_{\lambda\rho}$ acts by

$$q_i \mapsto h_i q_i.$$

On the orientation-preserving component, write

$$h_i = \exp(u_i + i\chi_i).$$

Then

$$A_i^\lambda \mapsto A_i^\lambda + du_i, \quad A_i^\rho \mapsto A_i^\rho + d\chi_i.$$

Curvature transforms invariantly in the abelian case:

$$F_i = dA_i \mapsto dA_i = F_i.$$

The Moebius component acts by orientation reversal on the ρ part and by the declared convention on the λ part.

9.3 Holonomy and Moebius closure

For a loop $\gamma \subset M$,

$$\text{Hol}_{\lambda\rho}(\gamma) = \mathcal{P} \exp \int_\gamma A_{\lambda\rho}$$

in the matrix case; in the abelian case,

$$\text{Hol}_{\lambda\rho}(\gamma) = \exp \int_\gamma (A_\lambda + iA_\rho).$$

If the loop crosses an orientation-reversing transition, then

$$\text{Hol}_{\lambda\rho}(\gamma) \in \mathbb{C}^\times \tau.$$

This is the formal Moebius condition. It is the non-metaphorical version of the claim that COTG self-return is not flat tautology.

10 COTG self-containment theorem-candidate

Definition 10.1 ($\alpha\rho$ generation operator). *Let $\Phi_{\alpha\rho}$ be the operation that takes a clock-declared hidden-support presentation, applies α -clock reconstruction/gluing together with ρ -polar transport, and returns the induced COTG presentation with its retained kernel, projection, and residual.*

Hypothesis 10.2 (COTG uniqueness / only geometry of hidden support). *Within the intended DA universe, any admissible geometry of hidden support factors through $\Phi_{\alpha\rho}$ and is therefore COTG up to declared equivalence.*

Theorem 10.3 (COTG self-containment, presentation form). *Assume the COTG uniqueness hypothesis and the $\lambda\rho$ COTG presentation equivalence theorem. Then COTG is a ρ -twisted fixed point of its own hidden-support geometry:*

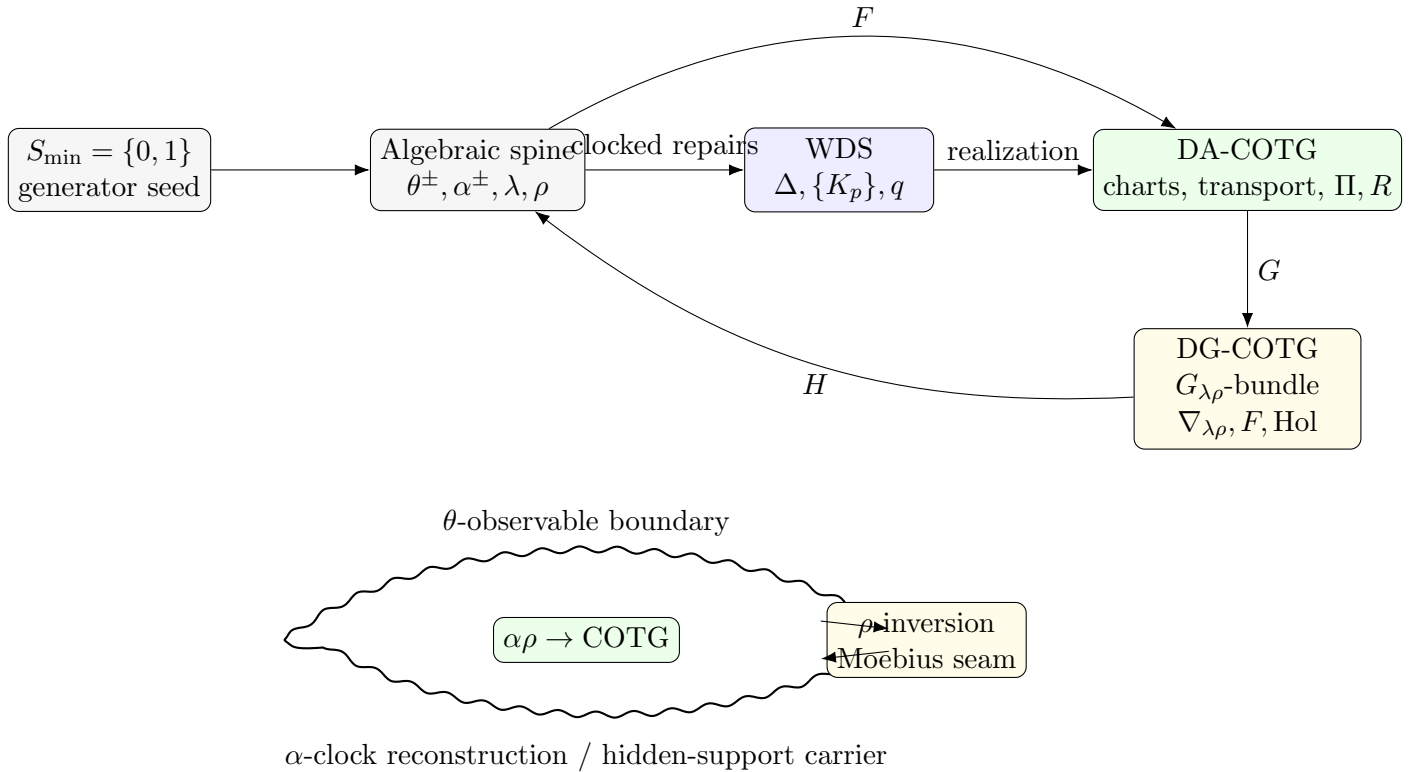
$$\boxed{\text{COTG} \simeq \Phi_{\alpha\rho}(\text{COTG}).}$$

The fixed point is non-tautological when the return monodromy includes polar/Moebius inversion.

Proof. By the uniqueness hypothesis, admissible hidden-support geometry factors through $\Phi_{\alpha\rho}$. Apply this to the hidden-support data carried by COTG itself: its charted fibers, retained kernels, transition maps, and residuals. This returns a COTG presentation. By the presentation equivalence theorem, the returned presentation is equivalent to the original one up to gauge and retained-kernel equivalence. If the return monodromy lies in the polar inversion component, the equivalence is not flat identity but ρ -twisted self-containment. $\square$

Remark 10.4 (Status). *This theorem is rigorous relative to the stated hypothesis and presentation category. The universal statement “COTG is the only geometry of hidden support” remains a hypothesis until the relevant universe of hidden-support geometries and admissible morphisms is fixed.*

11 Diagrammatic summary



12 Representation bridge to legacy mathematics

The representation bridge does not assert a naive bijection between θ -space and α -space. It asserts a typed representation equivalence: known legacy structures can be written from DA first principles, and DA-derived structures can be written in legacy mathematical language, provided the clocks, quotients, retained kernels, and residuals are declared.

The finite zeta-facing carrier is an example of this bridge. For a finite prime set P and exponent cutoff M , define

$$N_{P,M} = \{\nu = (\nu_p)_{p \in P} : 0 \leq \nu_p \leq M\},$$

$$T_{P,M} = \bigotimes_{p \in P} \text{span}_{\mathbb{C}}\{e_{p,0}, \dots, e_{p,M}\}, \quad E_{\nu} = \bigotimes_{p \in P} e_{p,\nu_p}.$$

The prime-coordinate exit map

$$\beta_{P,M}(\nu) = n_{\nu} = \prod_{p \in P} p^{\nu_p}$$

is bijective onto its finite support by unique factorization. No information is lost at this stage.

For $s = 1/2 + u + it$ define

$$\lambda_u(n) = e^{-u \log n}, \quad \rho_t(n) = e^{-it \log n},$$

and primitive binary residue

$$K_2(P) = \prod_{p \in P} \mu_2, \quad \kappa_P(n) = ((-1)^{v_p(n)})_{p \in P}.$$

The DA/COTG cell is

$$c_{P,M,s}(\nu) = (E_{\nu}, n_{\nu}, n_{\nu}^{-1/2} \lambda_u(n_{\nu}) \rho_t(n_{\nu}), \kappa_P(n_{\nu})).$$

The finite COTG state is

$$Q_{P,M}(s) = \sum_{\nu \in N_{P,M}} n_{\nu}^{-s} E_{\nu}.$$

The scalar projection

$$\Pi_{\Sigma}(E_{\nu}) = 1$$

gives

$$\Pi_{\Sigma} Q_{P,M}(s) = \sum_{n \in S_{P,M}} n^{-s}.$$

The bridge point is not a claim about RH. It is a clean example of representation grammar: tensor carrier, scalar label, λ -tilt, ρ -phase, and binary residue are retained before the non-invertible projection kills fiber labels.

13 Failure surfaces and proof obligations

13.1 Failure surfaces

Code	Failure surface
F1	Treating θ or α as objects instead of clocks.
F2	Using hidden support as data without a legal α -clock reconstruction channel.
F3	Declaring a projection without specifying its fiber and retained kernel.
F4	Collapsing a defect equivalence class without no-leakage.
F5	Treating COTG as the entire WDS rather than the admitted realization/quotient layer.
F6	Calling a self-loop tautological or non-tautological without computing its return monodromy.
F7	Claiming a global differential-geometric COTG while leaving transition maps, connection, or residual undefined.

F8	Claiming a bijection between θ -space and α -space rather than a typed representation equivalence.
F9	Treating the zeta-facing carrier as an RH proof rather than a representation-grammar bridge.
F10	Passing from finite to infinite tensor/completed objects without declaring topology and operator domains.

13.2 Proof obligations

1. Define the exact category of hidden-support geometries in which COTG is claimed to be unique.
2. Specify whether the Moebius inversion is conjugation, reciprocal conjugation, sign inversion, or a context-dependent involution.
3. Prove that every admissible hidden-support geometry admits $G_{\lambda\rho}$ transition functions.
4. Prove no-leakage for every quotient, especially any collapse of local binary residues into a global parity bit.
5. State the topology and completion used when finite COTG objects are replaced by infinite-dimensional or analytic completions.
6. Define morphisms in $\mathbf{Sp}_{\lambda\rho}$, **DA-COTG**, and **DG-COTG** tightly enough to promote presentation equivalence into a formal theorem.
7. Separate theorem, hypothesis, definition, and interpretation in future drafts.

14 Final formal definition

Definition 14.1 (Defective Analysis). *Defective Analysis is the clocked calculus of defective observables, hidden fibers, retained kernels, admissible reconstruction, and no-leakage. A DA instance assigns a clock signature*

$$\text{Clk}(P) \subseteq \{\theta, \alpha\},$$

constructs the θ -clock observable

$$D_\theta : X \rightarrow Y,$$

constructs the α -clock reconstruction

$$I_\alpha : Y \times K \rightarrow X / \sim,$$

and studies the induced fiber, retained kernel, quotient, polarity, admissibility rule, and residual:

$$\text{DA}(P) = (\text{Clk}(P), D_\theta, I_\alpha, F_\theta, K_{\text{ret}}, \Pi, \sigma, \text{Adm}, R).$$

Definition 14.2 (COTG). *Charted Oscillatory Transport Geometry is the $\lambda\rho$ -regular DA/WDS realization layer in which hidden-support reconstruction under α -clock operations is expressed as charted complex oscillatory transport with magnitude discipline, polar phase, gluing residuals, tensor aggregation, projection, and admissibility:*

$$\text{COTG} = (M, \mathcal{A}, E, \{X_i\}, \{q_i\}, \{g_{ij}\}, T, \Pi, R).$$

When upgraded to differential geometry, it is equivalently a $G_{\lambda\rho}$ -bundle geometry

$$(M, E_{\lambda\rho}, G_{\lambda\rho}, \nabla_{\lambda\rho}, \Pi, R, \text{Adm}), \quad G_{\lambda\rho} = \mathbb{C}^\times \rtimes \mathbb{Z}_2^\rho.$$

Definition 14.3 (Algebraic-spine derivation of COTG). *COTG is derived from the algebraic spine by taking the scalar closure*

$$\mathbb{C} = S_{\min} + \theta^{\pm} + \alpha^{\pm} + \lambda + \rho,$$

reading θ and α as clocks, reading λ as invariant magnitude/completion, reading ρ as polar phase/inversion, passing through WDS defect sectors and admissible reconstruction, and realizing the resulting $\lambda\rho$ transport as a charted differential geometry.

Definition 14.4 (Bijection / equivalence statement). *Under finite, non-leaking, clock-declared, $\lambda\rho$ -regular hypotheses, the algebraic spine presentation, the DA-derived COTG presentation, and the differential-geometric COTG presentation are equivalent:*

$$\boxed{\mathbf{Sp}_{\lambda\rho} \simeq \mathbf{DA-COTG} \simeq \mathbf{DG-COTG}.}$$

This is the formal meaning of the claimed bijection.

References and source basis

This draft synthesizes the uploaded June 2026 Defective Analysis documents:

1. *DA_01_Framing_Terminology*: clock assignment, primitive terms, problem-description protocol.
2. *DA_02_Defective_Calculus*: defective observables, retained kernels, admissible reconstruction.
3. *DA_04_COTG_From_DA*: WDS path and finite COTG realization.
4. *DA_Addendum_Global_Definitions*: WDS, primal calculus, clock-coordinate doctrine, generator seeds.
5. *Defective Analysis & the Numbers*: algebraic spine and λ, ρ scalar repairs.
6. *DA/COTG-Zeta Stipulation*: lossless pre-zeta carrier and typed projection-loss map.
7. *Defective Analysis: A Unified Internal Treatment*: consolidated corrections and geometry/COTG chapters.